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Final Project Proposal

1. A major real-world business problem that this agentic AI system addresses is managing customer service in real time within the restaurant industry. Restaurants operate in fast-paced and extremely competitive environments where efficiency, speed, and customer satisfaction are significant to success. Customer demand is often unpredictable, and during busy hours, staff members may struggle to respond to all customer inquiries while also juggling other duties.

    This pressure can lead to longer wait times, order mistakes, and displeased customers. Although these problems can be solved by simply hiring additional workers, it requires much more financial investment, training, and management efforts. The added responsibilities can strain restaurant operations.

    An agentic AI system offers a more effective alternative by handling customer service tasks, such as answering questions, providing menu recommendations, and filtering out options, as well as assisting with order placement based on the customer needs. By allowing the customers to interact with the AI agent, employees can focus on food preparation and in-person service. Unlike humans, the system can operate consistently and can handle large amounts of requests simultaneously. As a result, the system improves efficiency, reduces overall costs, and enhances customer experience!

2. There are many instances in which the agentic AI system can support both customers and restaurant staff. One common use case involves customers with dietary restrictions or personal preferences. For example, a customer who does not eat meat may use the AI agent to filter the menu and display only vegan/vegetarian options. The system will remove dishes with meat and highlight much more suitable alternatives. This reduces the time staff spend answering inquiries and allows customers to quickly find appropriate meals.

Another use case involves customers seeking recommendations. For instance, a customer might ask about the restaurant's signature or most popular dish. As a response, the system can access internal ordering data, analyze which menu items are ordered most frequently, and generate a personalized menu. By using real-time data, the AI agent can provide accurate and up-to-date information.

In both scenarios, the agentic AI improves efficiency by delivering fast and reliable responses. It also decreases staff workload and satisfies customers by giving consistent and personalized support.

3. The agentic AI system will use a multi-agent structure in which multiple specialized agents collaborate to provide efficient and accurate customer service. Each agent will be responsible for a specific task, making sure the system is operated in organized coordination.

   At the center, the manager agent uses LLM API, such as OpenAI's API to take care of communication between the user and other agents. This agent receives the customer's question, determines what task it is, and assigns the request to the appropriate agent.

   The menu information agent will be responsible for gathering and organizing the menu through the restaurant database, such as ingredients, prices, allergies, and dietary categories. This agent specifically confirms the information is accurate and up-to-date about available dishes for the customers.

   The recommendation agent analyzes customer preferences, ordering trends, and historical sales data to generate personalized menu suggestions. For instance, if the customer requests a signature dish, the agent will identify and recommend based on their needs. The menu's UI will change in real-time with the use of Figma, an interface design tool.

Lastly, the order support agent will assist the customer in placing and editing orders. The system verifies the item availability, calculates totals, and reviews the responses for clarity before sending the finalized results to the restaurant's internal ordering system.

Altogether, the agents will form a collaborative workflow in which specific tasks are divided, processed, and delivered in an efficient and accurate manner during high-demand periods. Version control such as GitHub will allow the system to run smoothly in the long term.

4. Just like any other agentic AI system, the structure may run into potential challenges and risks. The development of generative AI is still recent, so many problems arise despite its efficiency and accuracy.

   One major risk is hallucination, where the agent might generate incorrect or misleading information about menu items, ingredients, or pricing. To reduce this risk, the system will rely on structured and verified restaurant data. Another challenge could be context limitation, which may cause the system to lose track of user preferences during long conversations. However, the issue can be addressed by maintaining short-term memory.

   Several constraints might impact system usage, including API rate limits, privacy concerns, and computation limits such as processing speed and server capacity. Customer data will be protected based on data protection guidelines.

   Despite the challenges, the system will remain built as a functional prototype by following a structured timeline, beginning with the system design (1-2 weeks), followed by its development and multi-agent integration (2-3 weeks), and concluding with testing, evaluation, and preparation of its final presentation and report (2-3 weeks). This is all an estimation and may take an additional week on each implementation.